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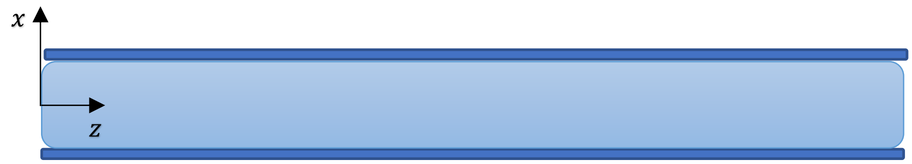
PROBLEM SET 4

TRANSPORT PHENOMENA I CHE 311

Transient laminar flow in a narrow slit

A Newtonian fluid of constant viscosity μ and constant density ρ is initially at rest between two large horizontal plates a distance $2B$ apart.



At time $t = 0$ a constant pressure gradient $\frac{\Delta p}{L} = \frac{p(z=L)-p(z=0)}{L}$ is imposed on the system, which sets the fluid in motion. Here, we study how the transient velocity profile develops with time.

1- Using the relevant equation of motion, show that the velocity profile is governed by the following PDE.

$$\rho \frac{\partial v_z}{\partial t} = \mu \frac{\partial^2 v_z}{\partial x^2} - \frac{\Delta p}{L}$$

2- Using no-slip boundary conditions, demonstrate that the steady-state velocity profile in the above system is as follows:

$$v_{z\infty} = v_{max} \left[1 - \left(\frac{x}{B}\right)^2\right]$$

, where $v_{max} = -\frac{B^2\, \Delta p}{2\mu L}$

3- Demonstrate that the partial differential equation derived in part 1 can be expressed in a dimensionless form as:

$$\frac{\partial \phi}{\partial \tau} = \frac{\partial^2 \phi}{\partial \eta^2} + 2$$

, where $\phi \equiv \frac{v_z}{v_{max}}$, $\tau = \frac{\mu t}{\rho B^2}$, and $\eta \equiv \frac{x}{B}$

4- Show that the steady-state velocity profile in dimensionless form is $\phi_\infty = 1 - \eta^2$.

5- Write down the initial condition and two boundary conditions for $\phi = \phi(\eta, \tau)$

$$\phi(\eta, \tau = 0) = \cdots$$
$$\phi(\eta = -1, \tau) = \cdots$$
$$\phi(\eta = +1, \tau) = \cdots$$

6- With the above boundary conditions, ϕ can be expanded in terms of cosine functions as:

$$\phi(\eta, \tau) = \sum_{n=1}^{\infty} C_n(\tau) \cos\left(\frac{2n-1}{2}\pi\,\eta\right)$$

Substituting the above summation in the partial differential equation $\frac{\partial \phi}{\partial \tau} = \frac{\partial^2 \phi}{\partial \eta^2} + 2$, derive an ordinary differential equation for $C_n(t)$ and evaluate the integration constants using the proper orthogonality equation for the cosine functions.

7- Plot ϕ as a function of η for $\tau = 0.01, 0.05, 0.1, 0.5, 1.0, 2.0,$ and $3.0.$

1) $V_z = V_z(x, t)$ $\quad \frac{\partial p}{\partial z} = \frac{\Delta p}{L}$

$$\rho\left(\frac{\partial v_z}{\partial t} + \cancel{v_x \frac{\partial v_z}{\partial x}} + \cancel{v_y \frac{\partial v_z}{\partial y}} + \cancel{v_z \frac{\partial v_z}{\partial z}}\right) = -\frac{\partial p}{\partial z} + \mu\left[\frac{\partial^2 v_z}{\partial x^2} + \cancel{\frac{\partial^2 v_z}{\partial y^2}} + \cancel{\frac{\partial^2 v_z}{\partial z^2}}\right] + \cancel{\rho g_z} \qquad \text{(B.6-3)}$$

$$\boxed{\rho \frac{\partial V_z}{\partial t} = \mu \frac{\partial^2 V_z}{\partial x^2} - \frac{\Delta p}{L}}$$

2) Steady state:

$$\mu \frac{\partial^2 V_{z\infty}}{\partial x^2} = \frac{\Delta p}{L}$$

$$\frac{\partial V_{z\infty}}{\partial x} = \frac{\Delta p}{\mu L} x + C_1 \qquad V_{z\infty} = \frac{\Delta p}{2\mu L} x^2 + C_1 x + C_2$$

$V_{z\infty}(x=\pm B)=0$ B.C

$$0=\frac{B^2\Delta p}{2\mu L}+BC_1+C_2$$

$$\nearrow 0=\frac{B^2\Delta p}{\mu L}+2C_2$$

$$0=\frac{B^2\Delta p}{2\mu L}-BC_1+C_2$$

$$\underline{C_2=-\frac{B^2\Delta p}{2\mu L}} \text{ y } \underline{C_1=0}$$

$$V_{z\infty}(x)=\frac{\Delta p}{2\mu L}x^2-\frac{B^2\Delta p}{2\mu L}=\frac{B^2\Delta p}{2\mu L}\left(\frac{x^2}{B^2}-1\right)$$

$$\boxed{V_{z\infty}(x)=V_{max}\left(1-\left(\frac{x}{B}\right)^2\right)}$$

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3) $\phi=\dfrac{v_z}{v_{max}}$ $\qquad d\phi=\dfrac{dv_z}{v_{max}}$ $\qquad dv_z=v_{max}\,d\phi$

$\tau=\dfrac{\mu}{\rho B^2}t$ $\qquad d\tau=\dfrac{\mu}{\rho B^2}dt$ $\qquad dt=\dfrac{\rho B^2}{\mu}d\tau$

$\eta=\dfrac{x}{B}$ $\qquad d\eta=\dfrac{dx}{B}$ $\qquad dx=B\,d\eta$

$$\frac{d^2v_z=v_{max}\,d^2\phi}{dx^2=B^2\,d\eta^2}$$

$$\rho \frac{\partial v_z}{\partial t} = \mu \frac{\partial^2 v_z}{\partial x^2} - \frac{\Delta p}{L} \qquad v_{max} = \frac{-B^2 \Delta p}{2\mu L}$$

$$\cancel{\rho} \frac{(v_{max}) \partial \phi}{\frac{\cancel{\rho} B^2}{\mu} \partial \tau} = \mu \frac{(v_{max}) \partial^2 \phi}{B^2 \partial \eta^2} - \frac{\Delta p}{L}$$

$$\frac{\partial \phi}{\partial \tau} = \frac{\partial^2 \phi}{\partial \eta^2} - \frac{\Delta p B^2}{\mu L v_{max}} \qquad \frac{-\Delta p B^2}{\mu L} \cdot \frac{2\mu L}{-B^2 \Delta p} = 2$$

$$\boxed{\frac{\partial \phi}{\partial \tau} = \frac{\partial^2 \phi}{\partial \eta^2} + 2}$$

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4) $v_{z\infty}(x=\pm B) = 0 \rightarrow \phi_\infty(\eta = \pm 1) = 0$ $\qquad \frac{\partial^2 \phi_\infty}{\partial \eta^2} = -2$

redimensionalizing!

$$\frac{\partial \phi_\infty}{\partial \eta} = -2\eta + C_1$$

$$\phi_\infty = -\eta^2 + C_1 \eta + C_2$$

$$\begin{cases} 0 = -1 + C_1 + C_2 \\ 0 = -1 - C_1 + C_2 \end{cases}$$

$$\hookrightarrow 0 = -2 + 2C_2 \rightarrow \underline{C_2 = 1} \rightarrow \underline{C_1 = 0}$$

$$\boxed{\phi_\infty = 1 - \eta^2}$$

5) $\phi(\eta, \tau=0) = 0$ } I.C

$\phi(\eta=-1, \tau) = 0$

$\phi(\eta=1, \tau) = 0$ } B.Cs (no-slip)

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6) Let $\phi = \phi_{\infty} + \phi_t$

$\underline{\phi_t(\eta=-1, \tau) = \phi(\eta=1, \tau) = 0}$

At $\tau=0$:

$0 = 1-\eta^2 + \phi_t(\eta, \tau=0)$ $\quad \underline{\phi_t(\eta, \tau=0) = \eta^2 - 1}$

$$\phi_t(\eta, \tau) = \sum_{n=1}^{\infty} C_n \cos(\lambda_n \eta) \qquad \lambda_n = \frac{2n-1}{2}\pi$$

$$\frac{\partial C_n}{\partial \tau} \cos(\lambda_n \eta) = -\lambda_n^2 C_n \cos(\lambda_n \eta)$$

$$\underline{\frac{\partial C_n}{\partial \tau} = -\lambda_n^2 C_n} \rightarrow \int \frac{1}{C_n} dC_n = \int -\lambda_n^2 \, d\tau$$

$$\ln|C_n| = -\lambda_n^2 \tau + B_n$$

$$\underline{C_n = B_n e^{-\lambda_n^2 \tau}}$$

From B.Cs:

$$\sum_{n=1}^{\infty} B_n \cos(\lambda_n \eta) = \eta^2 - 1$$

$\eta^2 - 1$ is even

$$\frac{a}{\pi} \int_0^{\frac{\pi}{a}} \cos(nax)\cos(max)\,dx = \delta_{nm}$$

$a = \pi$

$$\int_{-1}^{1} \cos^2(\lambda_n \eta)\,d\eta = 1 \quad , m = n$$

$$\int_{-1}^{1} (\eta^2 - 1)\cos(\lambda_n \eta)\,d\eta = B_n$$

$$\int_{0}^{1} (\eta^2 - 1)\cos(\lambda_n \eta)\,d\eta = \frac{B_n}{2}$$

$$B_n = \frac{-8}{(2n-1)^3 \pi^3}$$

$$C_n = \frac{-8}{(2n-1)^3 \pi^3} e^{-\lambda_n^2 \tau}$$

$$\phi_t = \sum_{n=1}^{\infty} \frac{-8}{(2n-1)^3 \pi^3} \cos(\lambda_n \eta)\, e^{-\lambda_n^2 \tau}$$

$$\boxed{\phi = (1-\eta^2) + \sum_{n=1}^{\infty} \frac{-8}{(2n-1)^3 \pi^3} \cos(\lambda_n \eta)\, e^{-\lambda_n^2 \tau}}$$

7)